NTv2_Vel Grid File Format

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The original NTv2 grid shift file format was modified for use with 3D velocity data. The term "grid shift" used in the original NTv2 documentation referred to coordinate shifts between two coordinate reference frames. However, one may use any type of grid data in place of coordinate shifts. For the NAD83_Vel velocity grid we have used 3D velocities in the local geodetic (north, east, up) coordinate system in place of horizonal coordinates shifts used in the original NTv2 format.

The new NTv2_Vel format is identical to the original NTv2 format except that (1) each data record was expanded to include a vertical component, and (2) the order of the data components were switched so that the velocities are given in the order velocity east (east +ve), velocity north (north +ve) and velocity up (up +ve). Instead of 4 elements per record for the two horizontal coordinate shift components and their accuracies, there are 6 elements per record for the three velocity components and their accuracies.

Like NTv2 grid files, the NTv2_Vel grid file consists of 11 Overview Header Records, followed by one or more Sub-Grid blocks, and terminating with an End of File Record. Each Sub-Grid block consists of 11 Sub-File Header Records followed by a series of Grid Data Records.

For each Overview and Sub-Grid Header record, the first data element is a header element name which describes the data in the second data element. Each Grid Data record contains six data elements (three velocity components and three corresponding accuracy components).

Note that in the Overview Header records, there are "from" and "to" reference system records, each with their semi-major and semi-minor axes records. Because the velocities are in a single reference system, the "from" and "to" reference system records will be identical.

There is a Grid Velocity record for each point (node) in the sub-grid. The first three data elements are the values of the three velocity components at the grid point in the order east, north and up. The next three data elements are the corresponding accuracies of the three velocity values, also in the order east, north and up. The number of Grid Data Records is given by:

$$\text{Grid Points Per Row} = \frac{\text{Upper Longitude} - \text{Lower Longitude}}{\text{Longitude Grid Interval}} + 1$$

$$\text{Grid Points Per Column} = \frac{\text{Upper Latitude} - \text{Lower Latitude}}{\text{Latitude Grid Interval}} + 1$$

$$\text{Number of Grid Records (n)} = \text{Grid Points Per row} \times \text{Grid Points Per Column}$$

The grid file is a Fortran 77 unformatted, direct-access binary file with a fixed record length of 24 bytes. An ASCII version of the format may be created and converted to binary format on other platforms.

The following describes each of the different records in a NTv2_Vel grid file.

Overview Header Records

Record #	*First Data Element*	*Second Data Element*
1	'NUM_OREC' Character*8	Number of Overview Header Records Integer*4
2	'NUM_SREC' Character*8	Number of Sub-Grid Header Records Integer*4
3	'NUM_FILE' Character*8	Number of Sub-Grids Integer*4
4	'GS_TYPE' Character*8	Grid Shift Data Type (units for velocities) Character*8
5	'VERSION' Character*8	Version ID of Velocity Model Character*8
6	'SYSTEM_F' Character*8	Reference System From Character*8
7	'SYSTEM_T' Character*8	Reference System To Character*8
8	'MAJOR_F'	Semi Major Axis From
9	'MINOR_F' Character*8	Semi Minor Axis From Real*8
10	'MAJOR_T' Character*8	Semi Major Axis To Real*8
11	'MINOR_T' Character*8	Semi Minor Axis To Real*8

NOTE: Corresponding “From” and “To” records will be identical.

Sub-Grid Header Records

Record #	*First Data Element*	*Second Data Element*
1	'SUB_NAME' Character*8	Sub-Grid Name Character*8
2	'PARENT' Character*8	Parent Grid Name Character*8
3	'CREATED' Character*8	Creation Date Character*8
4	'UPDATED' Character*8	Last Revision Date Character*8
5	'S_LAT' Character*8	Lower Latitude Limit of Grid (sec of arc +ve north) Real*8
6	'N_LAT' Character*8	Upper Latitude Limit of Grid (sec of arc +ve north) Real*8
7	'E_LONG' Character*8	Lower Longitude Limit of Grid (sec of arc +ve west) Real*8
8	'W_LONG' Character*8	Upper Longitude Limit of Grid (sec of arc +ve west) Real*8
9	'LAT_INC' Character*8	Latitude Grid Interval (sec of arc) Real*8
10	'LONG_INC' Character*8	Longitude Grid Interval (sec of arc) Real*8
11	'GS_COUNT' Character*8	Number of Grid Data Records Integer*4

Grid Velocity Records

Velocity Data Elements (6 per n grid node records)

Record # 12 ... n+11

East_Vel	North_Vel	Up_Vel	East_Acc	North_Acc	Up_Acc
Real*4	Real*4	Real*4	Real*4	Real*4	Real*4

End of File Record

Record #	*First Data Element*	*Second Data Element*
n+12	'END'	0.33e33
	Character*8	Real*8